TRAY D BRANCH SR.

Fremont, CA 94539
310-486-1012
tray.d.branch@gmail.com
https://www.linkedin.com/in/traydbranch/
https://traybranch.carrd.co/

PROFESSIONAL SUMMARY

AI and Systems Engineer with 10+ years of experience designing and deploying data-intensive and performance-critical systems across enterprise environments. Proven ability to build scalable software solutions integrating machine learning, computer vision, and real-time data processing. Experienced in developing end-to-end systems, from architecture and implementation to testing and optimization, using Python and modern ML frameworks, with growing proficiency in C++ for performance-oriented applications.

Strong background in distributed systems, data pipelines, and system reliability, with a track record of improving efficiency and enabling high-impact decision-making through engineered solutions. Military veteran with a disciplined, systems-driven approach to problem solving, reliability, and execution in mission-critical environments.

EDUCATION

Master of Science, Computer Science; California Science and Technology University – Dec. 2026
Bachelor of Science, Game Engineering; University of Silicon Valley – May. 2026
Bachelor of Science, Finance; East Carolina University – Dec. 2016
Associate of Science, General Studies; Wake Technical Community College – May. 2015

PROFESSIONAL EXPERIENCE

Accenture – Mountain View, CA **2020 – Present**
Senior Management Consultant – Data & AI (Client-Facing Consulting Role)

- Designed and implemented data-driven systems, including analytics platforms and scalable data pipelines, to support enterprise decision-making
- Translated business requirements into technical architectures for data and machine learning systems, enabling reliable and scalable solution deployment
- Collaborated with cross-functional engineering teams to integrate AI and automation solutions into production environments
- Optimized data processing workflows, improving system efficiency, performance, and reporting accuracy across client systems
- Delivered end-to-end solutions in fast-paced, client-facing environments, balancing system reliability, performance, and evolving requirements

Wells Fargo – Raleigh, NC **2018 – 2020**
Financial Analyst – Middle Market Banking

- Built and maintained financial modeling systems to evaluate asset performance, risk exposure, and cash flow sustainability
- Analyzed large datasets to assess credit risk, liquidity, and leverage, supporting lending and underwriting decisions
- Translated complex financial data into structured reports and actionable insights for stakeholders

Credit Suisse – Morrisville, NC **2017 – 2018**
Financial Analyst – Interest Rates Product Control

- Developed and maintained reporting systems for daily profit and loss, balance sheet validation, and risk verification
- Automated financial reporting processes to improve accuracy and reduce operational overhead
- Collaborated with trading and operations teams to analyze system outputs and ensure data integrity across financial systems

United States Army – Anchorage, AK **2010 – 2014**
System Administrator

- Supported and maintained distributed IT infrastructure serving 3,000+ users across mission-critical environments
- Managed large-scale system migrations and ensured high system availability and reliability
- Implemented security protocols and system-level procedures to protect infrastructure and maintain operational integrity

PROJECTS

PRISM: Predictive Perception Engine

- Designed and implemented a predictive perception system using deep learning to generate real-time attention maps, reducing rendering workload while maintaining visual fidelity
- Optimized data processing pipeline for perceptual prioritization, enabling efficient resource allocation in performance-constrained environments
- Integrated model inference with system-level decision logic to dynamically adjust rendering behavior based on predicted user focus

Aeroscapes: UAV Semantic Segmentation

- Developed and evaluated multiple deep learning architectures (U-Net, DeepLabv3+, SegFormer) for aerial image segmentation, achieving 0.789 mIoU across 12 classes
- Built a full data processing and training pipeline for large-scale image datasets (3,000+ samples), optimizing preprocessing and model evaluation workflows
- Analyzed model performance trade-offs and selected architectures based on accuracy and computational efficiency

Image Processing Framework

- Engineered a custom image processing framework in Python using OpenCV and NumPy, implementing edge detection, noise reduction, and compositing algorithms
- Designed modular processing components to handle pixel-level transformations efficiently under varying input conditions
- Optimized image restoration techniques to improve output quality in degraded signal environments

Robotics Systems (In Progress)

- Developed autonomous navigation logic in simulation environments (Webots), implementing sensor-driven obstacle avoidance and real-time decision-making
- Designed kinematic models for robotic arm control using numerical methods, focusing on forward and inverse kinematics computation
- Built modular robotics architecture using ROS 2 for distributed perception and control across system components

Backend/Systems

- Developing a backend service using FastAPI to process and respond to real-time input data, simulating scalable system behavior under concurrent requests
- Implementing API-driven architecture with input validation, processing logic, and structured output handling